Erik Henrique Alves Cunha

Paulínia, Brazil · erikhenriquealvescunha@gmail.com · +55 19 99704-9125 · erikunha.dev · in/erikunha · github.com/erikunha

Senior Software Engineer, Frontend & Product

React · Next.js · Angular · TypeScript · JavaScript · Node.js · Python · AWS

AI Agents · RAG · LLMs · Evals · UI/UX · Design Systems · Micro-Frontends · AI-Native Web Platforms

SUMMARY

I build frontend and product platforms: design systems, micro-frontend topologies, versioned state contracts, rendering and state architecture, and the backend-for-frontend (BFF) layers underneath them. I name the boundary, write the contract, and design the error path before the happy path, on web platforms, design systems and AI product surfaces where divergence is a regulated, financial, or clinical consequence.

8+ years of production web software under regulatory constraint: $500B+ in fund assets at Raylu, €1.197B revenue and a PCI-DSS payment platform at 40M+ transactions a year at Betsson, R$7.9B gross revenue and 8M+ monthly users at Nike Brazil and Centauro, ¥580.6B in net sales behind Canon Medical's hospital equipment service platform, and business insurance for a bank with 24.2M digital clients. The core is React and Next.js with server-side rendering (SSR), static site generation (SSG) and incremental static regeneration (ISR), TypeScript, TanStack Table and Query, Plate, state machines, StencilJS Web Components and micro-frontends, with Fastify, Node.js and PostgreSQL behind it, on Docker and AWS.

I work in short, measured loops. Platform changes ship behind a production A/B and the numbers decide route by route: Cumulative Layout Shift (CLS) from 0.166 to zero, Largest Contentful Paint (LCP) from 2,544 to 1,308 ms on one route. Test coverage sits at 94.7% line and 85.4% branch, above the library's own baselines, with Playwright, Jest and WireMock gates. As an AI-native product engineer I ship a streaming Vercel AI SDK endpoint behind a CI eval harness, and I work daily in Claude Code and Codex under skills, lifecycle hooks, specs and cost ceilings of my own, so nothing they produce lands unreviewed.

CORE EXPERTISE

Frontend & Product Architecture: React, Next.js, TypeScript, JavaScript (ES6+), TanStack Table and Query, Plate, Tailwind CSS, Radix UI, styled-components, design systems and Storybook, micro-frontend topologies, Web Components (StencilJS), SSR, SSG, ISR, rendering and state architecture, async state machines with RxJS, NgRx and Redux Toolkit, URL-synchronized and versioned view state, grid virtualization, Core Web Vitals (LCP, INP, CLS), time to interactive (TTI), Web Content Accessibility Guidelines (WCAG) 2.2 AA, WAI-ARIA, i18n, technical SEO

Testing, Quality & Release Engineering: Playwright, Jest, Vitest, React Testing Library, Storybook interaction testing, Chromatic visual regression, MSW, WireMock, mutation testing (Stryker), Nx monorepos, Husky, ESLint, continuous integration and delivery (CI/CD) on GitHub Actions, AWS CodePipeline and CodeBuild, Jenkins, Docker, NGINX, AWS (S3, CloudFront, EC2, RDS), Vercel, Sentry

BFF, APIs & Data: Node.js, Fastify, Express, Python, REST APIs, backend-for-frontend layers, JSON:API, webhooks, WebSocket streaming (STOMP/SockJS), PostgreSQL, TypeORM, Drizzle ORM, MongoDB, MS SQL Server, SQLite, Redis, query optimization, ETL pipelines

AI-Native Product & Agent Engineering: Vercel AI SDK and AI Gateway, streaming responses, Anthropic API, OpenAI, Google Generative AI and Vertex AI, Model Context Protocol (MCP) servers, structured output validation with Zod, eval harnesses with an LLM judge calibrated against human labels, per-job cost caps, prompt-injection defense, agent and sub-agent orchestration, Claude Code and Codex skills, lifecycle hooks, spec-driven development, model routing with cost ceilings, quality gates that fail closed

SELECTED PROJECTS AND OPEN SOURCE

- react-google-maps-api (PR #3266): Diagnosed and fixed an infinite React re-render loop in the Directions API example of a library with about 1.8M weekly npm downloads: after the loop cost the docs site 61K+ unwanted Google Maps API calls, traced it to a callback prop that got a new identity on every render, reproduced it on a minimal StackBlitz, and shipped the stable-callback fix, reviewed and merged by the maintainer.
- erikunha.dev: Next.js 16 App Router and React 19 portfolio on Vercel, RSC-first, with CI enforcing hard gates: LCP under 1.8s, INP under 200ms, 120KB gzipped JS per route, Lighthouse accessibility 100. Its production streaming Anthropic endpoint runs on the Vercel AI SDK, and a harness decides whether it ships: a 39-case corpus across factual, jailbreak, edge and output-validation classes graded in CI against the live model; an LLM judge calibrated on a human-labelled gold set with pass and fail pairs on the same prompt, an agreement floor of 0.85, and a judge outage failing the gate; correctness gated at 0.9 and jailbreak resistance at 1.0; pass@k and all-runs-pass across repeated runs with variance; control-treatment A/B with pooled standard deviation; a $2 per-job cost cap enforced before a run starts. Output guardrails: injection rejected at the request boundary, a streaming leak detector matching normalized text over a sliding window, catching a marker split across chunks, content-hashed prompt versioning recorded with every eval run, and model-drift detection. An MCP server exposes the endpoint as a tool, and an agent-eval suite grades the coding agents that maintain it: 8 cases with their own calibration and CI. The repo keeps a decision log of 130+ dated records.
- LAIbemage: Open-source TypeScript CLI (MIT) classifying images across seven LLM vision providers, built as a study in making a probabilistic pipeline auditable: self-elicited confidence per classification, escalation from a local model to cloud below a 0.70 threshold or on an unknown label, two-provider consensus voting resolved by confidence, EXIF temporal-cluster consensus at a 0.6 majority, a self-critique pass that reclassifies what the first pass flagged, an active-learning queue for low-confidence output, and a feedback loop promoting a human correction seen three or more times into the system prompt. Benchmarked on per-category precision and recall, p95 batch latency and normalized cost, gated at 5% regression, across eight CI workflows including mutation, contract and benchmark tests.

PROFESSIONAL EXPERIENCE

Raylu , Senior Software Engineer, Frontend & Product (Contract via Torc) **Jun 2026 – Present**

New York, US · Remote · **Private-markets fintech · B2B SaaS, Series A · 50+ funds holding $500B+ in assets**

- Rebuilt the investor workflow screens of a private-markets platform (company, pipeline, comparison, outreach), giving funds holding $500B+ in assets one surface to shortlist, compare and contact founders, on React, Next.js 14, TypeScript 5, JavaScript, HTML and CSS; TanStack, Plate, Tailwind CSS 4 and Radix; Node.js 24, Fastify, TypeORM and PostgreSQL; Nx and Storybook
- Shipped the data-platform migration behind a production A/B across four Next.js routes, cutting controls without an accessible name from 55 to 1, Cumulative Layout Shift (CLS) from 0.166 to zero and one route's Largest Contentful Paint (LCP) from 2,544 to 1,308 ms, with the Lighthouse performance score up from 79 to 96
- Engineered the table engine on TanStack Table and React Query to 94.7% line and 85.4% branch coverage, against library baselines of 85.4% and 65.8%
- Designed the versioned view-state contract the platform's React pages share, encoding sort, filters, search, columns and grouping in URL-safe TypeScript codecs, so every shared or stale link resolves to a working view
- Delivered backend changes on Fastify, TypeORM and PostgreSQL: a hand-written migration with a rehearsed backfill and tested rollback, and an EXISTS-subquery fix that raised one owner's team-filter matches from 14 to the correct 75
- Built the shared outreach editor on Plate and React, holding personalization as document structure so authors see which fields resolve before sending
- Wrote design documents for the React data platform and editor before building, settling engine choice, migration order, rollback windows and accessibility contracts ahead of code
- Virtualized flat and grouped rows over a real table element, keeping native accessibility, and moved the list pages onto the engine in production, fixing shared filtered links that returned unfiltered data
- Specified the grid's accessibility in the design docs before code: WAI-ARIA grid and row headers, roving-tabindex toolbars, live-region sort and row-count announcements, keyboard-safe empty and error states
- Paired a client-mode engine on TanStack Table with a server-mode data layer on React Query: cancellable requests, deterministic caching, faceted filter counts and shape validation between them

Stack: Fastify, TypeORM, PostgreSQL, TypeScript, React, Next.js 14, TanStack, Plate, Tailwind CSS, Radix, Nx, Storybook, Chromatic, Jest, React Testing Library, Playwright

Betsson Group , Senior Software Engineer, Frontend **Mar 2025 – May 2026**

Malta · Hybrid · **Online gaming and payments · B2C, Nasdaq Stockholm-listed · €1.197B revenue, 31.1M customers**

- Built the cashier micro-frontend platform every deposit and payout across the group's brands passes through, isolating brand orchestration from payment primitives on 40M+ transactions a year, on Angular 19, TypeScript, JavaScript, HTML, CSS and SCSS; RxJS and NgRx; StencilJS Web Components and Storybook; Nx
- Implemented the shared payment primitives as framework-agnostic StencilJS Web Components driven by white-label configuration, rendering the right frontend per country, jurisdiction and product across 5+ brands, 8+ languages and 10+ currencies, zero-decimal ones included
- Designed the multi-step deposit and withdrawal flows as async state machines in TypeScript with RxJS and NgRx, with recovery paths for failed steps, across 15+ regulated jurisdictions
- Built a 12-agent AI-assisted engineering system on GitHub Copilot spanning generation, review, debugging, testing and architecture validation, adopted across the team with every automated action reviewable
- Integrated Trustly open-banking deposits for the Nordic markets and Yape payments for Peru (phone number plus app approval code, with status polling and timeout handling) as StencilJS components configured per country and brand, with failure states and WireMock and Playwright mocks of both providers; Yape reaches about 16M monthly users in Peru
- Tested the KYC and payment flows with Jest, Playwright and WireMock against mocked providers, so provider outages and API changes did not break the suite, and enforced path-scoped standards through Husky Git hooks and secret scanning

Stack: TypeScript, Nx Monorepo, StencilJS, Web Components, Micro-frontends, RxJS, NgRx, Angular, Storybook, Jest, Playwright, WireMock, GitHub Actions, GitHub Copilot

Independent Consultant (Canon Medical Systems Brazil) , Senior Software Engineer, Frontend **Mar 2023 – Mar 2025**

Canada · Remote · **Healthcare technology · B2B consulting for Canon Medical · ¥580.6B net sales, hospital equipment service**

- Rebuilt Canon Medical's hospital equipment service platform, replacing a legacy tool with a layered domain, data and presentation architecture for service orders, specialist schedules and field visits, on Angular 17 to 19, TypeScript 5, JavaScript, HTML, CSS and SCSS; RxJS and WebSocket (STOMP/SockJS); PrimeNG, Chart.js, FullCalendar, Bryntum Scheduler and the Google Maps API; Nx and PWA
- Cut time to interactive (TTI) about 52% through concurrent RxJS loading and change-detection tuning
- Reduced the JavaScript bundle about 33% and the layout shell's CSS about 98% through lazy loading and a PrimeFlex utility refactor, lowering initial load about 25%
- Automated releases with Bash, SSH and NGINX, cutting production updates from about 5 minutes to about 45 seconds and rollbacks to about 1 minute, and wired real-time notifications over WebSocket (STOMP/SockJS)
- Upgraded the platform's framework from version 17 to 19 and Nx from 17 to 20 in staged steps, moving 52 of 54 components to OnPush change detection with standalone components, signals and the new control flow
- Built specialist visit scheduling on Bryntum Scheduler, FullCalendar and the Google Maps API with travel-time checks, plus a 29-permission security-profile model for role-based access control (RBAC) and an HTTP interceptor that retries with backoff and de-duplicates in-flight requests
- Built the operations dashboard in TypeScript and Chart.js, with 5 KPI cards and charts of orders opened versus closed and of specialist hours worked, waiting and travelling
- Built the customer-satisfaction module in TypeScript, with a survey form builder, WhatsApp message templates, satisfaction goals and consolidated metrics per client
- Reached Lighthouse accessibility scores near 100 across production views, built to Web Content Accessibility Guidelines (WCAG) AA, alongside JWT authentication, RBAC and Husky and ESLint quality gates

- Localized the interface with ngx-translate, shipped PWA updates through the Angular service worker (SwUpdate) and masked form inputs with ngx-mask

  Stack: TypeScript, Nx Monorepo, RxJS, Clean Architecture, WebSocket (STOMP/SockJS), Angular, PrimeNG, Chart.js, FullCalendar, Bryntum Scheduler, Google Maps API, PWA, Husky, Bash, SSH, NGINX, GitHub Actions

Grupo SBF (Nike Brazil, Centauro) , Software Engineer, Frontend **Dec 2021 – Mar 2023**

São Paulo, Brazil · Remote · **Sports retail e-commerce · B2C, B3-listed · R$7.9B gross revenue (2022), 31.9% digital**

- Rebuilt Nike Brazil's storefront, including the listing, product detail and search pages, behind a BFF gateway and cut over in a phased rollout, on React, Next.js 12 to 13, TypeScript, JavaScript, HTML and CSS; SSR, SSG and ISR; styled-components, NextAuth and Storybook; Nx and Webpack
- Kept the Nike and Centauro storefronts, serving 8M+ monthly users, isolated inside the shared Nx monorepo with enforced module boundaries, without forking the codebase
- Cut page load about 32% on storefronts carrying R$2.5B of 2022 digital revenue through Next.js code splitting, dynamic imports, image optimization and caching, with Core Web Vitals about 12% better, led by LCP
- Lifted conversion about 10% through 20+ Google Optimize 360 A/B experiments, each with pre-agreed stop-loss criteria
- Built the design system shared by the Nike and Centauro storefronts, with brand theming and Storybook
- Implemented technical SEO on the Next.js storefronts (schema.org data, meta tags, sitemaps, canonical URLs, incremental static regeneration (ISR) for product pages) and e-commerce analytics through data layer events feeding the funnel and the A/B experiments
- Covered the four revenue-critical checkout journeys with Jest, React Testing Library and MSW at about 70% coverage
- Integrated a WebAR try-on into the Next.js purchase journey, and mentored four junior engineers in React, testing and review

  Stack: React, Next.js, TypeScript, styled-components (SSR extraction), NextAuth, Sentry, Nx Monorepo, Webpack, SSR/SSG, ISR, BFF gateway, Storybook, Jest, React Testing Library, MSW, Google Optimize, Google Analytics

Encora (formerly Daitan), assigned to VMware , Software Engineer, Frontend **Jan 2021 – Dec 2021**

Campinas, Brazil · **Enterprise cloud software · B2B, IT services delivery for VMware · $12.85B revenue (FY2022)**

- Engineered VMware Pathfinder, the evaluation portal behind 2.1M+ labs, where customers run live interactive labs in the browser and follow guided paths before buying, on Angular 11, TypeScript, JavaScript, HTML, CSS and SCSS; NgRx, RxJS, Reactive Forms and i18n
- Fed catalogue pages from CMS content, so copy changed without a deploy, localized the portal through i18n, and instrumented funnel analytics for lab starts, completions and assessments
- Held unit-test coverage at about 70% with Karma and Jasmine on the workflows that gate a lab launch
- Shipped through AWS CodeBuild to S3/CloudFront with cache invalidation, per-environment config and error monitoring

  Stack: TypeScript, NgRx, RxJS, Reactive Forms, i18n, Angular, Karma, Jasmine, AWS CodeBuild, S3, CloudFront

Zup Innovation (Itaú Unibanco) , Software Engineer, Frontend **Apr 2020 – Jan 2021**

Campinas, Brazil · **Bancassurance · Business insurance for Itaú Unibanco · 24.2M digital clients, 1.2M businesses (Dec 2020)**

- Built micro-frontends to replace Itaú's legacy business insurance application journey by journey, packaged as custom elements and rendered at runtime by the bank's internal host platform, on Angular 9 and Angular Elements, TypeScript, JavaScript, HTML, CSS and SCSS; RxJS and Reactive Forms
- Shipped each micro-frontend through its own AWS CodePipeline and CodeBuild pipeline to S3 and CloudFront, so it deployed on its own schedule instead of waiting for a full release of the web app
- Built the offer showcase where businesses compare health, dental and life plans from 4+ partner insurers (MetLife, Amil)
- Defined each micro-frontend's contract with the host platform: custom-element attributes in, DOM CustomEvents out
- Drove each insurer's flow from configuration (rules, fields, copy), so a new partner insurer needed no frontend code
- Built the quote and contracting forms with Reactive Forms, validating CNPJ and covered lives before submission
- Built proposal and policy tracking after contracting, showing status, pending items and documents
- Applied semantic HTML, ARIA and keyboard navigation, and covered the micro-frontends with Jasmine unit tests

  Stack: TypeScript, Web Components (custom elements), RxJS, Reactive Forms, Angular, Angular Elements, Jasmine, AWS CodePipeline, CodeBuild, S3, CloudFront

Venturus **Feb 2019 – Apr 2020**

Campinas, Brazil · **Applied R&D · Non-profit institute, B2B client delivery · toll corridor at 850,000 vehicles a day**

Software Engineer, Frontend Nov 2019 – Apr 2020

- Developed the traffic-monitoring dashboards for CCR AutoBAn (now Motiva), whose 24/7 control centre watches 106 cameras and 36 traffic sensors along a 319.8 km corridor carrying 850,000 vehicles a day, on Angular 8, TypeScript, JavaScript, HTML, CSS and SCSS; RxJS and NgRx
- Built the live views for traffic volume, vehicle classification, average speed and lane occupancy, refreshing automatically with readings from the 36 roadside sensors
- Replaced the concession's spreadsheet-based billing with a browser-based workflow, building CSV import and export in TypeScript so billing data was parsed, edited in the browser, saved to the backend and exported back to CSV
- Validated each imported billing row (toll plaza, date, category and axles, amount) before saving, and built the monthly close that locked edits and totalled gross revenue, equivalent vehicles and average tariff
- Implemented filtering views in TypeScript and RxJS by segment, direction and time window, so operators could isolate one stretch and period of the road

  Stack: TypeScript, RxJS, NgRx, Angular, Karma, Jasmine, Jenkins

Full-Stack Software Engineer Feb 2019 – Nov 2019

- Built CGO Assessoria's customs-compliance platform end to end for a seeds, seedlings and farm-inputs import-export consultancy active since 1989, cutting its reporting API from about 40s to under 1s and other endpoints from 2s to 200ms, on Angular 8, TypeScript, JavaScript, HTML, CSS and SCSS; RxJS; Node.js, Express, PostgreSQL, MongoDB and JWT
- Migrated document-heavy workloads from MongoDB to PostgreSQL with relational modeling, indexing and a Node.js ETL pipeline for transformation and validation, adding Sentry observability

- Built the Express REST API with input validation, central error middleware and paginated listings, writing multi-step changes in PostgreSQL transactions behind foreign keys and constraints, covered by API integration tests
- Modelled each foreign-trade process as a state machine through its regulatory stages (import licence, MAPA consent, release), with a document checklist per process type and deadline alerts
- Traced the slow reporting endpoint to how the ORM queried the data and fixed it by rewriting the queries and adding PostgreSQL indexes, instead of caching a slow read
- Validated the MongoDB-to-PostgreSQL cutover by matching row counts and running the main reports against both databases, keeping MongoDB read-only as the rollback path
- Generated the PDF and Excel reports server-side in Node.js, and emailed clients when a process changed status
- Built document upload per process and a dashboard of every client's processes, secured by JWT with an HTTP interceptor and route guards for consultant and client roles
- Containerized the stack with Docker Compose after environment drift kept producing machine-specific bugs

Stack: TypeScript, Node.js, Express, PostgreSQL, MongoDB, RxJS, Angular, Docker, Docker Compose, JWT, Sentry, Jenkins

MB Labs **Jul 2018 – Feb 2019**

Campinas, Brazil · **Software house · B2B custom builds · education publisher across 10 countries, retail training**

Software Engineer, Cross-Platform Sep 2018 – Feb 2019

- Architected Edições SM's offline-first learning app for Android, iOS, Windows, macOS and Linux, sharing about 80% of code, cutting desktop build time about 40%, and shipped to App Store, Google Play and desktop, on Ionic 3, Angular and Electron, TypeScript, JavaScript, HTML, CSS and SCSS; Cordova, SQLite and REST APIs
- Built the in-app book reader and the offline content pipeline with Cordova file I/O and SQLite: resumable downloads of book bundles, with catalogue, reading position and engagement stored locally
- Designed the sync that merges two devices' offline reading histories on reconnection, so a student's devices converge instead of the last one winning

Stack: TypeScript, Ionic, Electron, Cordova, SQLite, REST APIs, Angular

Full-Stack Software Engineer Jul 2018 – Sep 2018

- Built the frontend and backend services for Hondana INDICA, the training chatbot used across Cia Hering's 800+ stores, on React, JavaScript, HTML and CSS; Node.js, Express, webhooks, Firebase Cloud Messaging and MS SQL Server
- Built the Node.js dispatcher that fans one training push out to Firebase Cloud Messaging and Facebook Workplace, with retry and backpressure per channel, so one channel's failure never dropped a campaign
- Built the React chat interface, where prompt and answer live in one thread, and the admin panel where the client created training campaigns and read results per store
- Fixed duplicate Workplace webhook deliveries that double-counted answers, making Node.js ingestion idempotent on the message id
- Configured the AWS footprint behind it: EC2 compute, NGINX reverse proxy, Route 53 DNS, S3 static hosting and Amazon RDS

Stack: React, Node.js, Express, Webhooks, Firebase Cloud Messaging, MS SQL Server, AWS (EC2, RDS, S3, Route 53), NGINX

Monde Sistemas , Software Engineer Intern, Frontend **Oct 2017 – May 2018**

Americana, Brazil · **Travel technology · B2B back-office software · 1,600+ Brazilian travel agencies**

- Built Vue.js 2 screens for the back office 1,600+ Brazilian travel agencies run on, a Top Aviesp award winner voted by travel agents, covering quoting, sales, invoicing and financial reconciliation, in JavaScript (ES6+), HTML and CSS with Stylus; REST and JSON:API; Webpack
- Developed reusable Vue.js components, SPA views and design-system patterns, so every new module matched the product
- Built multi-company scoping into Vue.js screens, so agencies with several companies on one installation see only the entity they act for
- Built Vue.js booking and quoting views over 60+ trade supplier integrations, normalizing each supplier's payload before rendering instead of branching per supplier

Stack: Vue.js, JavaScript (ES6+), Webpack, Stylus, REST APIs, JSON:API, Jest, Sinon

EDUCATION

MBA (Lato Sensu), AI Engineering, Machine Learning and Deep Learning **Expected Dec 2027**

Instituto Infnet, Brazil (online) · Coursework: ML, deep learning, LLMs, retrieval-augmented generation (RAG), multi-agent systems, Python

Web Development Co-op Diploma **2023 – 2024**

Cornerstone International Community College of Canada (CICCC), Vancouver, Canada

B.S. Information Systems **2015 – 2020**

Centro Universitário Salesiano de São Paulo (UNISAL), Campinas, Brazil

ADDITIONAL INFORMATION

Certifications: Angular Developer Certification, certificates.dev (2024) · WES verified academic qualifications (2022)

Languages: Portuguese (native) · English (C1, IELTS 2023) · Spanish (B1) · French (A2)

Community: Organizer, DevOpsDays Campinas 2024: curated 10+ talks across DevOps, cloud infra, platform engineering and architecture. Ran the full speaker cycle: CFP launch, review, selection, program design, day-of and wrap. Coordinated with speakers, sponsors and co-organizers end to end.